New product introduction TRAVEO™ T2G Automotive solutions

Jan., 2025

TRAVEO™ T2G features

Infineon scalable platform solutions

- Deliver full range of silicon products per platform
- Enable OEM / Tier 1 SW reuse (consistent across platform, generations)
- Deliver best-in-class, auto-quality solution components
- Reduce risk: Infineon is an established Automotive Semiconductor supplier

Infineon Automotive solution architecture

System software (customer/partner)

Auto-quality software (for example, MCAL)

Entry silicon	Mid-range silicon	High-end silicon

Auto-quality IP blocks (for example: compute, connectivity, graphics, and storage)

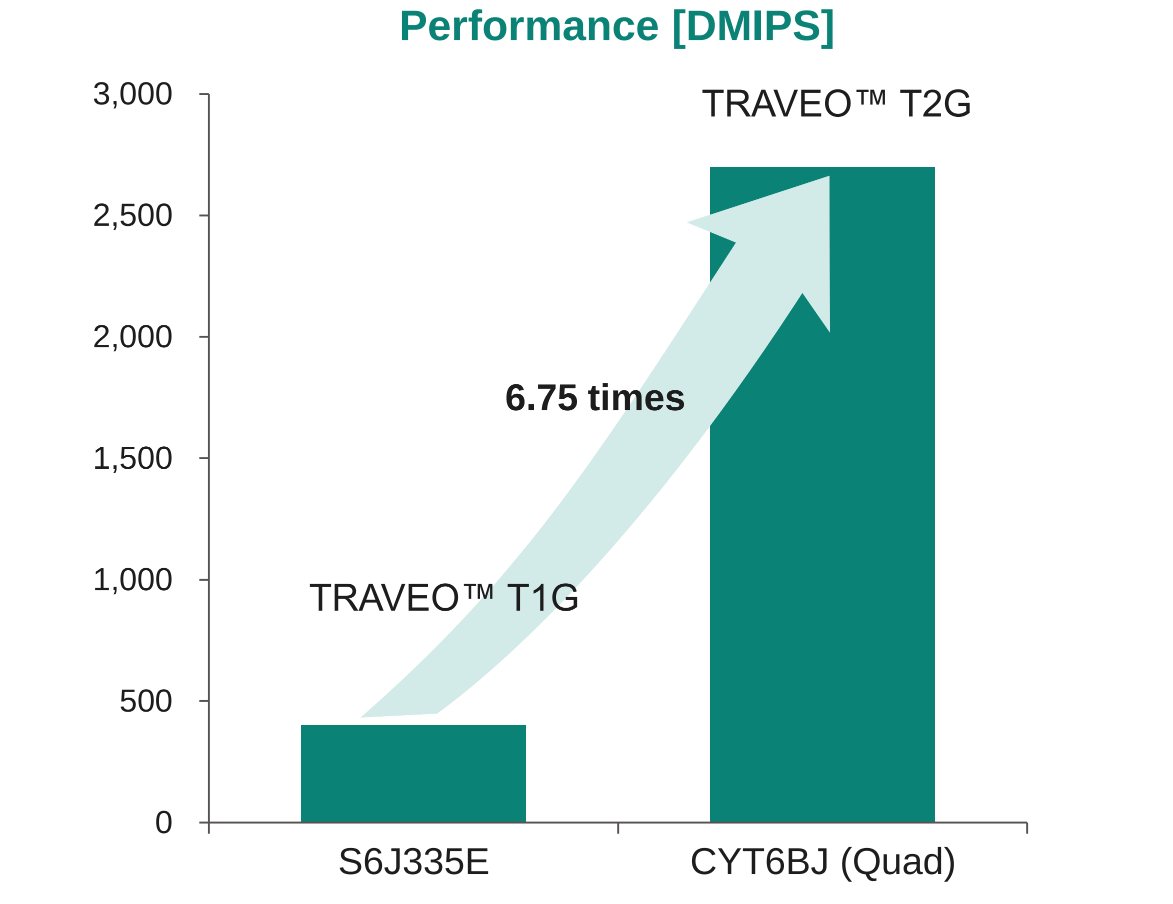
Key features TRAVEO™ T2G Body

Low-power	– Energy-efficient processing power
Performance	– Quadl Arm® Cortex®-M7 – 2700DMIPS
Scalability	– Complete portfolio – Memory density – Package lineup and performance
Safety	– ISO 26262 ASIL-B
Audio	– I^2S/TDM
Security	– Hardware security module: eSHE[1], HSM[2] – Evita full – ISO21434 ready
Connectivity	– LIN. CXPI – CAN FD – 1 Gb ethernet
Updatability	– FOTA[3] with RWW[4] flash – eMMC[5] – QSPI/HS-SPI

[1] eSHE: enhanced secure hardware extension
[2] HSM: Hardware security module
[3] FOTA: Firmware update over-the-air
[4] RWW: Read while write
[5] Embedded multimedia card

High-performance MCU

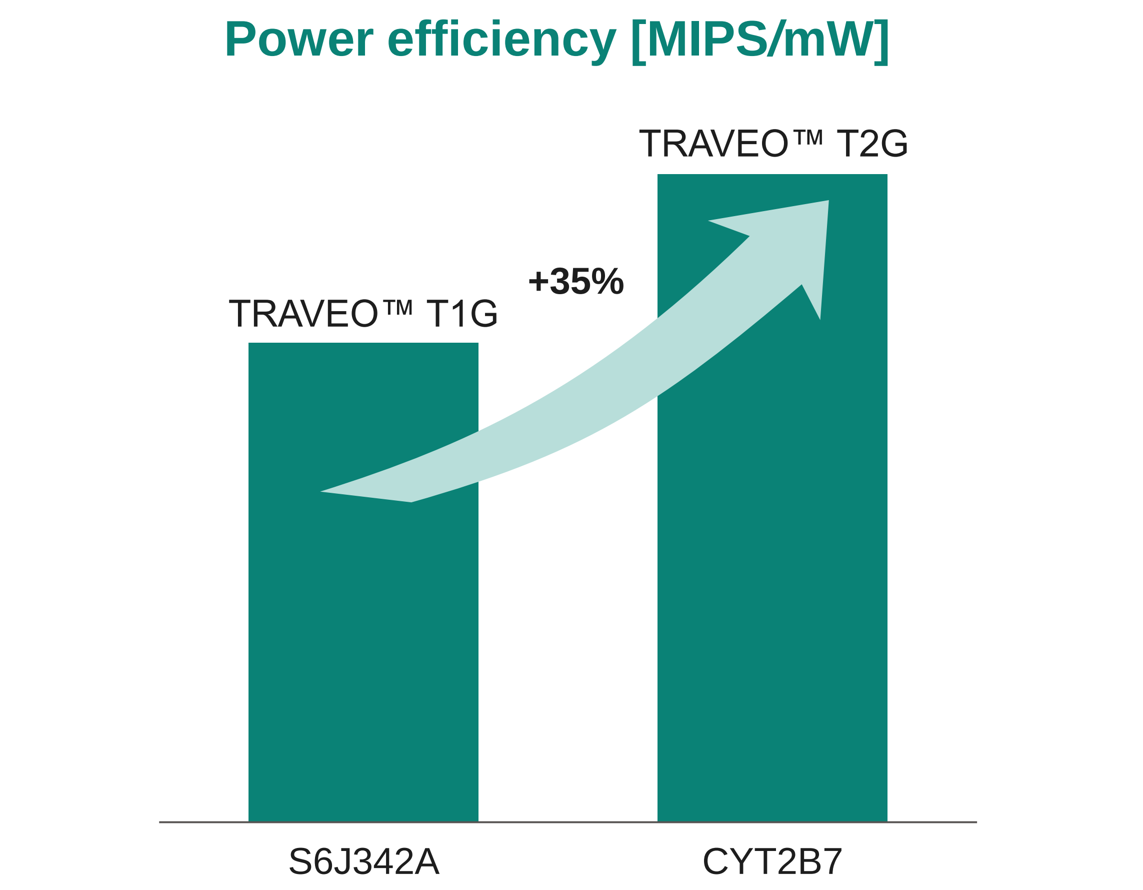
- Arm® Cortex®-M cores
  - Single M4 up to Quad M7*
  - Dedicated M0+ for security
  - Performance up to 2700 DMIPS*
    *: Up to Dual M7 and 1500 DMIPS for graphics products
- Hi-speed embedded flash with prefetch/cache
- Dedicated memory- and peripheral-DMA for CPU offloading
- Gigabit ethernet and CAN-FD for Hi-speed vehicle communication



TRAVEO™ T2G provides world-class performance

Power efficiency

- 35% improvement in power efficiency
- More power saving modes
  - Low-power Active
  - Sleep
  - Low-power Sleep
  - Deep Sleep
  - Hibernate
- Deep Sleep mode as low as 35 µA (typical)
- Hibernate mode as low as 5 µA (typical)



TRAVEO™ T2G achieves world-class energy efficiency

CAN, LIN, ethernet – automotive networks

Infineon supports state-of-the-art in-vehicle communication

CAN-FD

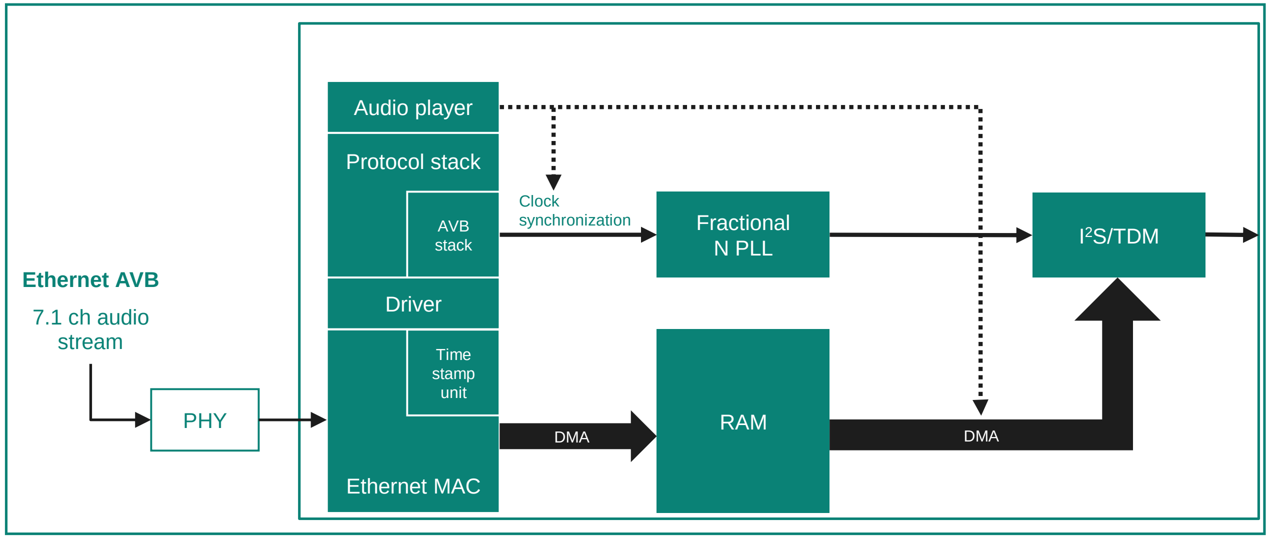
- Proven compliance to ISO11898-1 and ISO 16485
- implements the time-triggered CAN (TTCAN) protocol specified in ISO 11898-4 (TTCAN protocol levels 1 and 2) completely in hardware
- Maximum 8 Mbps supported
- Fully retained in Deep Sleep mode
- Shared message RAM with ECC protection
- DMA access to receive FIFOs

LIN

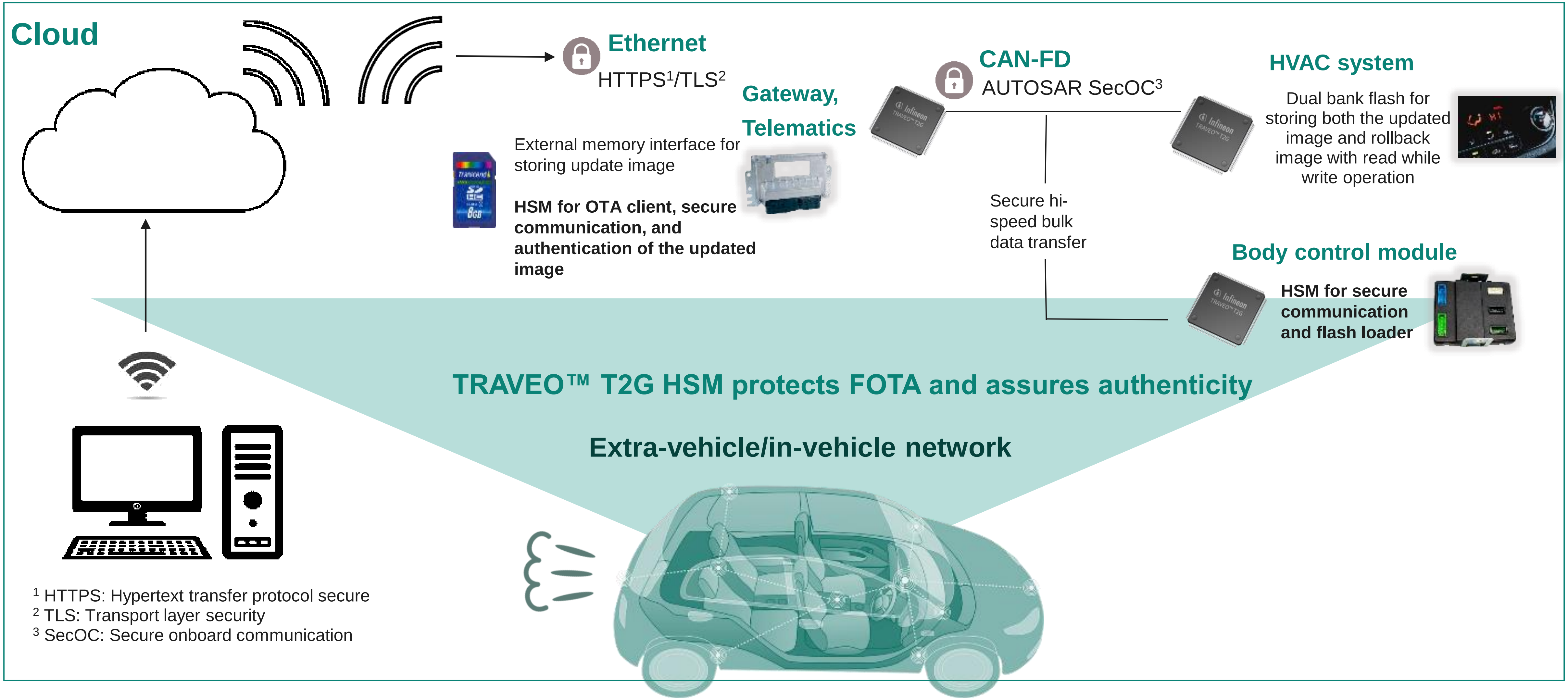
- LIN protocol support in hardware according to ISO 17987
- Master and slave functionality
- Autonomous header transmission/reception
- Autonomous response transmission and reception
- Message buffer for PID, data, and checksum fields

Ethernet

- 10/100/1000 Mbps ethernet MAC compatible with IEEE 802.3 and IEEE-1588 PTP
- Support of MII, RMII, and RGMII PHYs
- DMA interface
- Supports ethernet AVB and integrates fractional PLL for clock synchronization
- Supports full-duplex data transport using external PHY devices



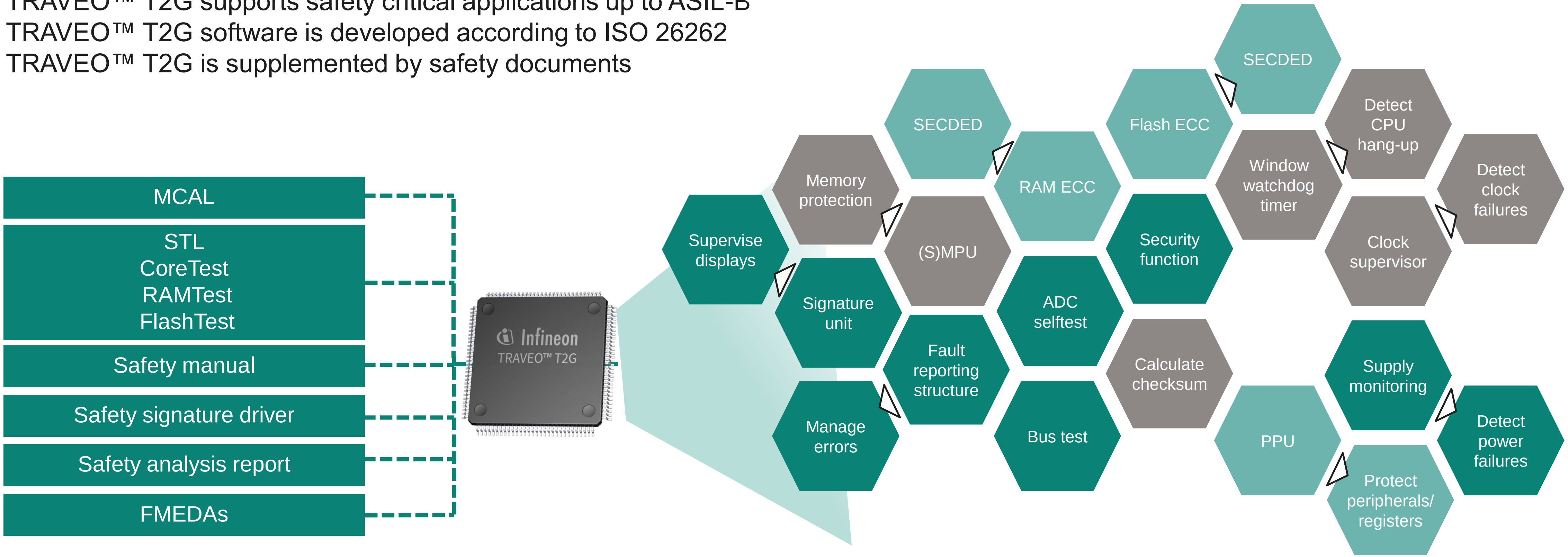
TRAVEO™ T2G use case: Firmware over-the-air (FOTA) update



[1] HTTPS: Hypertext transfer protocol secure
[2] TLS: Transport layer security
[3] SecOC: Secure onboard communication

Functional safety: A holistic system-level approach

- TRAVEO™ T2G is an ISO 26262 safety-element-out-of-context product
- TRAVEO™ T2G supports safety critical applications up to ASIL-B
- TRAVEO™ T2G software is developed according to ISO 26262
- TRAVEO™ T2G is supplemented by safety documents



TRAVEO™ T2G offers safety hardware, software, and documents

Functional safety with TRAVEO™ T2G

- Infineon provides the following support for enabling safe applications with TRAVEO™ T2G

HW safety manual

FMEDAs

for individual TRAVEO™ T2G products

SW products

including safety documentation

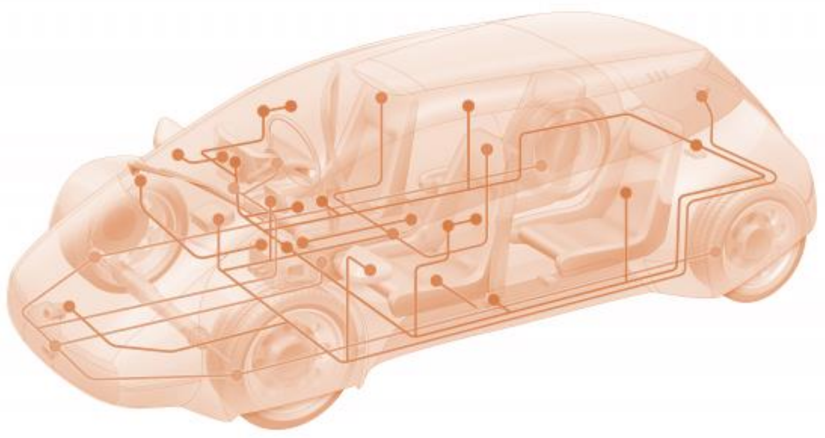
- These documents help to achieve functional safety at the system level
- Requirements have been derived to detect potential failure modes and to achieve the hardware architectural metrics for ASIL-B

Automotive security

ADAS, autonomous driving, and new digital service models require authentication and secure communication
TRAVEO™ T2G integrates HSM to support secure applications

Connected car at security risk

- Wiretapping
- Disguised identity
- Privacy/identity theft
- Unauthorized feature activation
- Unauthorized tuning
- Unlocking speed limit
- Forgery of driving record
- Hardware/property theft
- Manipulation of safety mechanism

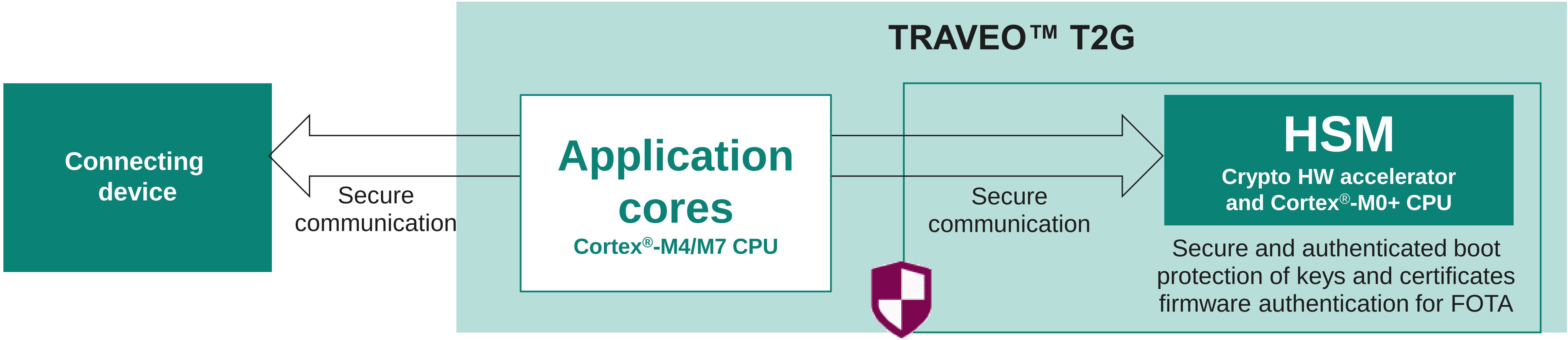


TRAVEO™ T2G keeps connected car secure

TRAVEO™ T2G HSM for security

- Root-of-trust boot ROM and chain-of-trust boot firmware
  - Ensure establishment of hardware isolation between secure and non-secure applications
  - Enable fast authentication of ECU software during secure boot

- Flexible configuration of secure domain for efficient resource utilization
- Generation and storage of device-unique secret AES keys



TRAVEO™ T2G product lineup

TRAVEO™ T2G body MCU portfolio

Body/gateway/infotainment

TRAVEO™ T2G Entry MCU	TRAVEO™ T2G High MCU

Peripherals-based on Infineon MXS40 platform

Performance

CYT2B6
Cortex®-M4, 80 MHz[1],
ASIL-B, HSM[2],
512 KB flash,
64 KB RAM
64 to 100[4] -LQFP

CYT2B7
Cortex®-M4, 160 MHz[1],
ASIL-B, HSM[2],
1 MB flash, 128 KB RAM
64 to 176[4] -LQFP

CYT2B9
Cortex®-M4, 160 MHz[1],
ASIL-B, HSM[2], CXPI[3]
2 MB flash, 256 KB RAM,
64 to 176[4]-LQFP

CYT2BL
Cortex®-M4,160 MHz[1],
ASIL-B, HSM[2], CXPI[3]
4 MB flash, 512 KB RAM
64 to176[4]-LQFP

CYT3BB/CYT4BB
2x Cortex®-M7[5], 250 MHz[1],
ASIL-B, HSM,
4 MB flash, 768 KB RAM,
eMMC, ethernet, I^{2}S/TDM[6]
100 to 176[4]-TQFP,
272-BGA[7]

CYT4BF
2x Cortex®-M7, 350 MHz[1],
ASIL-B, HSM,
8 MB flash, 1024 KB RAM,
eMMC, ethernet,
FlexRay[8], I^{2}S/TDM[6]
176[4]-TQFP,
272-BGA[7],
320-BGA

CYT6BJ Q125
4x Cortex®-M7,
320 MHz[1],
ASIL-B, HSM,
16 MB flash, 2048 KB RAM,
eMMC, ethernet,
FlexRay[8], I^{2}S/TDM[6]
176[4]-TQFP,
272-BGA[7],
320-BGA

1 Maximum operating frequency
2 Hardware security module, SHE-only mode available
3 Clock eXtension peripheral interface
4 Package pin count
5 Single or dual core option
6 Time division multiplexing
7 Fine pitch ball grid array
8 Optional feature

	Concept	Development	Sampling	Production
Status				
Availability			QQYY	QQYY

TRAVEO™ T2G Body portfolio scalability

Part number	Flash memory size	Pin count						
		LQFP/TQFP					BGA	
		64-pin	80-pin	100-pin	144-pin	176-pin	272-ball	320-ball
CYT6BJ	16 MB					2048 KB	2048 KB	2048 KB
CYT4BF	8 MB			High-end		1024 KB	1024 KB	1024 KB
CYT3BB/4BB	4 MB			768 KB	768 KB	768 KB	768 KB	
CYT2BL	4 MB	512 KB	512 KB	512 KB	512 KB	512 KB		Entry
CYT2B9	2 MB	256 KB	256 KB	256 KB	256 KB	256 KB		
CYT2B7	1 MB	128 KB	128 KB	128 KB	128KB	128 KB		
CYT2B6	512 KB	64 KB	64 KB	64 KB				

Legend: RAM

Common SW

Software offering

– MCAL[1]
– STL[2]
– FEE[3]

[1] MCAL: microcontroller abstraction layer
[2] STL: Self-test library
[3] FEE: flash EEPROM emulation

TRAVEO™ T2G for low-end to high-end Body ECUs

TRAVEO™ T2G Body MCU ordering code decoder

CY	T	X	X	X	X	B	X	A	Q	X	XX	X	GS	X
Company ID	ATV MCU family	MCU Core	Application	cFlash/wFlash/RAM density	Package	Hardware	Marketing	Revision	Fab Location	Reserved	Package code	Temperature	Quality	Tape/Reel shipment (opt.)

Company ID
- **CY** Infineon

ATV MCU family
- **T** TRAVEO™ T2G

MCU Core
- **2** Single core M4
- **3** Single core M7
- **4** Dual core M7
- **6** Quad core M7

Application
- **B** Body

cFlash/wFlash/RAM density
- **6** 512 KB/64 KB/64 KB (CYT2B6 series)
- **7** 1 MB/96 KB/128 KB (CYT2B7 series)
- **9** 2 MB/128 KB/256 KB (CYT2B9 series)
- **B** 4 MB/256 KB/768 KB (CYT3BB/CYT4BB series)
- **L** 4 MB/128 KB/512 KB (CYT2BL series)
- **F** 8 MB/256 KB/1024 KB (CYT4BF series)
- **J** 16MB/512KB/2048KB (CYT6BJ series)

Package
- **3** 64-pin
- **4** 80-pin
- **5** 100-pin
- **7** 144-pin
- **8** 176-pin
- **B** 272-ball
- **C** 320-ball

Hardware
- **B** eSHE
- **C** HSM (RSA 2K)
- **D** HSM (RSA 4K)

Marketing

	Ethernet	FlexRay	eMMC
A	off	off	off
B	1-ch	off	off
C	1-ch	on	off
D	1-ch	on	on
E	1-ch	off	on
F	2-ch	off	off
G	2-ch	on	off
H	2-ch	on	on
J	2-ch	off	on

Package code
- **AE** TQFP
- **AZ** LQFP
- **BZ** BGA

Temperature
- **S** -40 to +105°C
- **E** -40 to +125°C

Quality
- **GS** Qualified

Tape/Reel shipment (opt.)
- **Non-**Tray
- **T** Tape and reel shipment

TRAVEO™ T2G ecosystem

Comprehensive tools, kits, and software

Software

- Header files and sample driver libraries (SDL)
- AUTOSAR MCAL 4.2.x
- Self test libraries

Third-party software IDEs

- Green Hills Multi and IAR Embedded Workbench
- iSYSTEM debug and test environment und DTS development environment

Third-party debug hardware

- Green Hills and SuperTrace probe
- IAR I-jet debugging for Arm® Cortex®-M
- Lauterbach

Hardware

- Evaluation board
- Lite KIT

Functional safety

- Safety manual
- FMEDAs

Evaluation board

- Third-party HMI tools
- Third-party HSM software

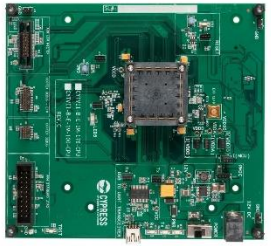
Other support from Infineon

- SPICE-verified software services and JTAG flash programming
- Auto Flash Utility

Extensive Infineon and partner development resources simplify system integration

TRAVEO™ extensive Kit ecosystem
Something for every design adventure and budget!

TRAVEO™ T2G Body Entry



- Develop and test the key functionalities provided by TRAVEO™ T2G such as User **Switch**, User **LED**, and **UART** communication
- **Socket** and **solded** version available

– CYTVII-B-E-100-SO
– CYTVII-B-E-176-SO
– CYTVII-B-E-1M-100-CPU
– CYTVII-B-E-1M-176-CPU
– CYTVII-B-E-2M-100-CPU
– CYTVII-B-E-2M-176-CPU
– CYTVII-B-E-4M-176-CPU

TRAVEO™ T2G Body High

- Develop and test functionalities such as **Audio Interface**, Automotive **Ethernet**, **SD Card**, **SMIF**, **Dual QSPI** User **Switch**, User **LED**, and **UART** communication
- **Socket** and **solded** version available

– CYTVII-B-H-176-SO
– CYTVII-B-H-272-SO
– CYTVII-B-H-320-SO
– CYTVII-B-H-4M-176-CPU
– CYTVII-B-H-4M-272-CPU
– CYTVII-B-H-8M-176-CPU
– CYTVII-B-H-8M-272-CPU
– CYTVII-B-H-8M-320-CPU
– CYTVII-B-H-16M-176-CPU
– CYTVII-B-H-16M-320-CPU

TRAVEO™ T2G Body Low cost kits

- **Low-cost**
- **Easy to use** evaluation board based on the TRAVEO™ T2G body **Entry/High** families **Ethernet**, **Arduino, mikroBUS**
- **Modustoolbox™ compatibility**

– CYTVII-B-E-1M-SK
– KIT_T2G-B-E_LITE ***new***
– KIT_T2G-B-H_LITE ***new***

TRAVEO™ T2G Lite kits supported by ModusToolbox™

TRAVEO™ T2G Lite kits
KIT_T2G-B-E_LITE
Fully supported by ModusToolbox™
KIT_T2G-B-H_LITE
Fully supported by ModusToolbox™

ModusToolbox™

ModusToolbox™ Software is a modern, extensible development ecosystem supporting a wide range of Infineon microcontroller devices, including **PSOC™ Arm® Cortex® microcontrollers**, **TRAVEO™ T2G Arm® Cortex® microcontroller**. Provided as a collection of development tools, libraries, and embedded runtime assets, ModusToolbox™ Software is architected to provide a flexible and comprehensive development experience

- User manual
- Getting started
- ModusToolbox™ product presentation
- GitHub
- Community support

TRAVEO™ T2G Preferred Design Houses

TRAVEO™ T2G preferred design houses

Our TRAVEO™ T2G preferred design houses is a trusted partners' ecosystem that extends the support force by tailoring their know-how to meet your specific needs.

By partnering with one of our qualified preferred design houses, you can be assured that you'll receive expert advice and customized support to help you achieve your goals. Our team of professionals brings added value to customer service, working together to optimize your design and help you succeed in your business objectives.

We understand that every customer is unique, which is why we offer tailored solutions to meet your specific needs. From product-specific support to application-specific advice, our preferred design house is fully trained to use TRAVEO™ T2G and provides a wealth of knowledge and expertise to help you succeed.

Together with our partners, we offer optimized customer support for systems using our products. Our preferred design houses are committed to delivering exceptional service and support to ensure your success

	Supported Products		Supported Region(s)					
PDH	TRAVEO™ Body	TRAVEO™ Cluster	Global	EMEA	AMR	JP	GC	AP
Altia	-	X	X	-	-	-	-	-
Avin Systems	-	X	-	X	-	-	-	X
Candera	-	X	X	-	-	-	-	-
Embedded Office	X	X	-	X	-	-	-	-
Embien Technologies	-	X	-	-	-	-	-	X
Elektrobit Automotive GmbH	X	X	X	-	-	-	-	-
G-pulse	-	-	-	-	-	-	X	-
Hightec	X	X	X	X	-	-	-	-
Hitex	X	X	X	X	-	-	-	-
L4B software	-	X	-	X	-	-	-	-
Macnica	X	X	-	-	-	X	-	-
Neutron Controls	X	X	-	-	X	-	-	-
QT	-	X	X	-	-	-	-	-
Revotech	X	X	-	-	-	-	-	X
Siili Auto	-	X	-	X	-	-	-	-
Techrein	X	X	-	-	-	-	-	X
Tekall	-	-	-	-	-	-	X	-

TRAVEO™ T2G software offering overview

AUTOSAR 4.2.2 SW (ASIL-B)

- MCAL[1]: MCU[2], ADC[3], ICU[4], GPT[5], PWM[6], WDG[7], OCU[8], CAN[9], LIN[10], SPI[11], FLS[12], DIO[13], and PORT
- STL (Self-test libraries): core test, flash test, and RAM test
- FEE (EEPROM emulation)
- Complex device drivers for I²C, UART, program flash
- Multi-core extension for MCAL, offering ASR 4.4 type II multi-core support for selected modules

Software services/customization

- Infineon SW teams have leading expertise in the fields: AUTOSAR, graphics, functional safety, and security
- Customized SW modules available upon request

SDL (sample driver library)

[1] MCAL: Microcontroller abstraction layer
[2] MCU: Microcontroller
[3] ADC: Analog digital converter
[4] ICU: Input capture unit
[5] GPT: General purpose timer
[6] PWM: Pulse width modulation
[7] WDG: Watchdog
[8] OCU: Output compare unit
[9] CAN: Controller area network
[10] LIN: Local interconnected network
[11] SPI: Serial peripheral interface
[12] FLS: flash
[13] DIO: Digital input/output

Compiler/programmer/debugger/probes for TRAVEO™ T2G

<table>
<tr><th>Vendor</th><th>SW tool</th><th>Compiler</th><th>Programmer</th><th>Debugger</th><th>ETM trace via SWD/JTAG[1])</th><th>Trace via TPIU (4 pins)</th><th>Debugger I/F</th></tr>
<tr><td>IAR</td><td>IAR Embedded Workbench</td><td>Yes</td><td colspan="3">I-jet</td><td>No</td><td>SWD/JTAG</td></tr>
<tr><td>IAR</td><td>IAR Embedded Workbench</td><td>Yes</td><td colspan="4">I-jet trace</td><td>SWD/JTAG/TPIU</td></tr>
<tr><td>Lauterbach</td><td>PowerView</td><td>No[2]</td><td colspan="4">µTrace</td><td>SWD/JTAG/TPIU</td></tr>
<tr><td>Lauterbach</td><td>PowerView</td><td>No[2]</td><td colspan="3">PowerDebug USB 3 + Cortex®-M debug cable
PowerDebug Pro + Cortex®-M debug cable</td><td>combined with CombiProbe</td><td>SWD/JTAG/TPIU</td></tr>
<tr><td>Green Hills Software</td><td>Multi</td><td>Yes</td><td colspan="3">Green Hills Probe</td><td>No</td><td>SWD/JTAG</td></tr>
<tr><td>Green Hills Software</td><td>Multi</td><td>Yes</td><td colspan="4">Green Hills SuperTrace Probe</td><td>SWD/JTAG/TPIU</td></tr>
<tr><td>iSYSTEM</td><td>WinIDEA</td><td>No[2]</td><td colspan="4">iC5000 / iC5700</td><td>SWD/JTAG/TPIU</td></tr>
<tr><td>Dts Insight</td><td>microVIEW-PLUS</td><td>Arm® DS MDK-ARM</td><td>adviceXross NETIMPRES</td><td colspan="3">adviceXross</td><td>SWD/JTAG/TPIU</td></tr>
<tr><td>PLS</td><td>UDE</td><td>No[2]</td><td>PLS-MemTool</td><td>UAD2pro/UAD2next/ UAD3+</td><td colspan="2">UAD2next/UAD3+</td><td>SWD/JTAG/TPIU</td></tr>
</table>

[1] Check with the tool vendor for the exact trace support feature and the latest status of handling of TVT2G MCUs
[2] Vendor does not offer own compiler

myInfineon collaboration platform for TRAVEO™ T2G

Access to additional technical documentation

By registering in the myInfineon collaboration platform (MyICP), you can get access to add-on technical documentation, trainings, tools, and much more for all TRAVEO™ T2G devices.

How to get access

If not already available, please create a myInfineon account on www.infineon.com. Please contact traveo@infineon.com and request access to TRAVEO™ T2G myICP.

Link to TRAVEO™ T2G MyICP

Infineon